Tutorial 09

ENVX2001 – Applied Statistical Methods

Semester 1

Welcome to week 9!

This week's lecture introduced you to how we use models for prediction and how we assess the quality of the predictions.We will follow similar steps in today's tutorial and lab class.

In this tutorial we will cover the key steps to fit a model, use it to make predictions, and then check assumptions, and select the most appropriate model.

You will learn how to:

1. Split data into training and test subsets
2. Fit your model to the training data and check assumptions are met
3. Use the trained model to make predictions using the training and the test subsets
4. Observe the quality of the predictions using appropriate statistics and plots

Packages

This tutorial uses `tidyverse`, `readxl`, `moments`, `psych`, `performance` and `car`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("tidyverse", "readxl", "moments", "psych", "performance", "car"))
```

If a package is already installed, R will simply reinstall it — no harm done.

Downloads

File	Used in	Download
`Dippers.csv`	all exercises	Download

Save the file into a folder called `data` inside your project folder.

The data

Figure 1: Cute little Dipper (bird) standing on a rock by the water. Image source: markmedcalf via Adobe Stock.

Dippers are thrush-sized birds living mainly in the upper reaches of rivers, which feed on benthic invertebrates by probing the river beds with their beaks. The dataset in this exercise contains data from a biological survey which examined the nature of the variables thought to influence the breeding of British dippers.

Twenty-two sites were included in the survey. Some of the variables have been transformed.

Data can be found in: **Dippers.csv**

The variables measured were:

- `Altitude` site altitude
- `Hardness` water hardness
- `RiverSlope` river-bed slope
- `LogCadd` the numbers of caddis fly larvae, transformed
- `LogStone` the numbers of stonefly larvae, transformed
- `LogMay` the numbers of mayfly larvae, transformed
- `LogOther` the numbers of all other invertebrates collected, transformed
- `Br_Dens` the number of breeding pairs of dippers per 10 km of river

In the analyses, the four invertebrate variables were transformed using a $log(x + 1)$ transformation.

`Br_Dens` will be the response variable in our models.

Last week following use of the step() function, we found the most parsimonious model to be:

Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogOther

Model assumptions were met and the model explained 76% of variation in dipper breeding pair density, which was quite good.

We will use this model in our walkthrough today.

Data splitting

Making predictions involves substituting in new values of x into our model equation to get new values of y.

We could use single values for each of our predictor variables and substitute them into the equation, but this is unrealistic and time consuming when we have a large number of observations. We therefore use a dataset containing our new observations of x to make the predictions and feed this into our model.

To understand the ability our model to predict new values of y from values of x previously unseen by the model, we need to test this on data that has not been used to fit the model.

It is common practice to split the data we have into two parts: a training (or calibration) subset and a test (or validation) subset. For each subset, we feed the predictor values into the model to get predictions of the y variable. We can then compare the predicted y with the observed y contained within each subset to get an indication of how well our model is performing.

Let's read the data in and start splitting!

CODE

```
# read in the data
dippers <- read.csv("data/Dippers.csv")

library(tidyverse)
glimpse(dippers) # check the data
```

OUTPUT

```
Rows: 22
Columns: 8
$ Altitude   <int> 259, 198, 251, 184, 145, 145, 198, 160, 251, 159, 160, 145,…
```

```
$ Hardness   <dbl> 12.20, 22.00, 26.30, 22.50, 29.50, 39.90, 42.80, 59.60, 69.…
$ RiverSlope <dbl> 10.90, 14.70, 6.90, 4.60, 1.91, 5.00, 6.20, 14.30, 4.60, 3.…
$ LogCadd    <dbl> 2.303, 2.890, 3.784, 4.419, 3.219, 3.932, 3.664, 4.431, 3.7…
$ LogStone   <dbl> 5.242, 4.344, 5.231, 5.242, 3.829, 4.898, 4.357, 6.337, 5.4…
$ LogMay     <dbl> 0.000, 3.401, 5.826, 5.749, 5.509, 5.749, 5.371, 0.000, 4.8…
$ LogOther   <dbl> 1.386, 1.609, 1.386, 1.386, 1.099, 3.045, 1.386, 2.944, 2.5…
$ Br_Dens    <dbl> 3.60, 4.30, 3.80, 3.40, 3.80, 4.50, 4.30, 5.00, 4.50, 3.40,…
```

There are different ways of splitting data, but what you choose generally depends on your dataset and the question you are trying to answer.

For example, we might split our dataset based on:

- a random selection of observations (random sampling)
- certain factors or groups within the dataset (stratified sampling)
- spatial location
- time

For today, we will be making a random split of the data, which is the most common method.

Because the sample size is very small, we will use 80% of the data for training and 20% for testing. The more observations we have to train the model, the better.

CODE

```r
set.seed(42)

indexes <- sample(1:nrow(dippers), size = 0.2 * nrow(dippers)) # randomly sample 20% of rows in
the dataset

dippers_train <- dippers[-indexes, ] # remove the 20% - training dataset
dippers_test <- dippers[indexes, ] # select the 20% - test dataset
```

In the code above, we first set a seed to ensure that we can reproduce the same random sample if we were to rerun the code chunk. Then we use the `sample()` function to randomly select 20% of the row indices from the dataset.

We create the training dataset by removing these selected indices from the original dataset ([-index,] says “exclude everything but these indices”), and we create the test dataset by selecting only those indices.

Once we have split the data, we need to check the subsets. First ensure the structure and number of observations in each subset is correct, and then check the distribution of the variables (max, mean, median, min) in each subset to ensure they are similar.

CODE

```r
str(dippers_train)
```

OUTPUT

```
'data.frame':   18 obs. of  8 variables:
 $ Altitude  : int  198 251 184 145 198 160 251 160 145 214 ...
 $ Hardness  : num  22 26.3 22.5 39.9 42.8 ...
```

```
 $ RiverSlope: num  14.7 6.9 4.6 5 6.2 14.3 4.6 15.8 5.9 5.7 ...
 $ LogCadd   : num  2.89 3.78 4.42 3.93 3.66 ...
 $ LogStone  : num  4.34 5.23 5.24 4.9 4.36 ...
 $ LogMay    : num  3.4 5.83 5.75 5.75 5.37 ...
 $ LogOther  : num  1.61 1.39 1.39 3.04 1.39 ...
 $ Br_Dens   : num  4.3 3.8 3.4 4.5 4.3 5 4.5 4.5 6.9 3.5 ...
```

CODE

```
str(dippers_test)
```

OUTPUT

```
'data.frame':   4 obs. of  8 variables:
 $ Altitude  : int  83 145 259 159
 $ Hardness  : num  153.9 29.5 12.2 68
 $ RiverSlope: num  2.88 1.91 10.9 3.3
 $ LogCadd   : num  2.83 3.22 2.3 4.45
 $ LogStone  : num  3.04 3.83 5.24 3.53
 $ LogMay    : num  5.9 5.51 0 6.03
 $ LogOther  : num  5.83 1.1 1.39 3.47
 $ Br_Dens   : num  2.9 3.8 3.6 3.4
```

CODE

```
summary(dippers_train)
```

OUTPUT

```
    Altitude        Hardness        RiverSlope       LogCadd     
 Min.   :107.0   Min.   : 15.50   Min.   : 4.60   Min.   :2.890  
 1st Qu.:160.0   1st Qu.: 40.62   1st Qu.: 5.75   1st Qu.:3.784  
 Median :187.5   Median : 84.73   Median : 7.25   Median :4.040  
 Mean   :195.8   Mean   : 74.07   Mean   :10.34   Mean   :4.326  
 3rd Qu.:236.5   3rd Qu.: 94.05   3rd Qu.:14.60   3rd Qu.:4.804  
 Max.   :305.0   Max.   :150.50   Max.   :26.20   Max.   :5.951  
    LogStone         LogMay         LogOther        Br_Dens     
 Min.   :3.434   Min.   :0.000   Min.   :1.099   Min.   :3.400  
 1st Qu.:4.599   1st Qu.:5.617   1st Qu.:1.848   1st Qu.:4.000  
 Median :5.175   Median :5.947   Median :3.132   Median :4.500  
 Mean   :5.215   Mean   :5.545   Mean   :3.419   Mean   :4.937  
 3rd Qu.:5.751   3rd Qu.:6.277   3rd Qu.:4.726   3rd Qu.:5.000  
 Max.   :6.791   Max.   :6.880   Max.   :6.773   Max.   :8.460  
```

CODE

```
summary(dippers_test)
```

OUTPUT

```
    Altitude        Hardness        RiverSlope        LogCadd     
 Min.   : 83.0   Min.   : 12.20   Min.   : 1.910   Min.   :2.303  
 1st Qu.:129.5   1st Qu.: 25.18   1st Qu.: 2.638   1st Qu.:2.700  
 Median :152.0   Median : 48.75   Median : 3.090   Median :3.026  
 Mean   :161.5   Mean   : 65.90   Mean   : 4.747   Mean   :3.202  
 3rd Qu.:184.0   3rd Qu.: 89.47   3rd Qu.: 5.200   3rd Qu.:3.528  
 Max.   :259.0   Max.   :153.90   Max.   :10.900   Max.   :4.454  
    LogStone         LogMay         LogOther        Br_Dens     
 Min.   :3.045   Min.   :0.000   Min.   :1.099   Min.   :2.900  
 1st Qu.:3.406   1st Qu.:4.132   1st Qu.:1.314   1st Qu.:3.275  
 Median :3.678   Median :5.705   Median :2.426   Median :3.500  
 Mean   :3.910   Mean   :4.360   Mean   :2.944   Mean   :3.425  
 3rd Qu.:4.182   3rd Qu.:5.933   3rd Qu.:4.056   3rd Qu.:3.650  
 Max.   :5.242   Max.   :6.031   Max.   :5.826   Max.   :3.800  
```

We can see that the training dataset contains 18 observations and the test dataset contains 4 observations, which is very small, so we don't expect to get very reliable information from the prediction quality statistics, but it still worth continuing to understand the process and see how we go.

Because of the limited sample size, we can see that the max and min values vary, but the distributions aren't too different (e.g. we don't have our response in one subset following a log distribution and the other following a normal distribution.)

From here we can proceed to model fitting.

Fit the model to the training data

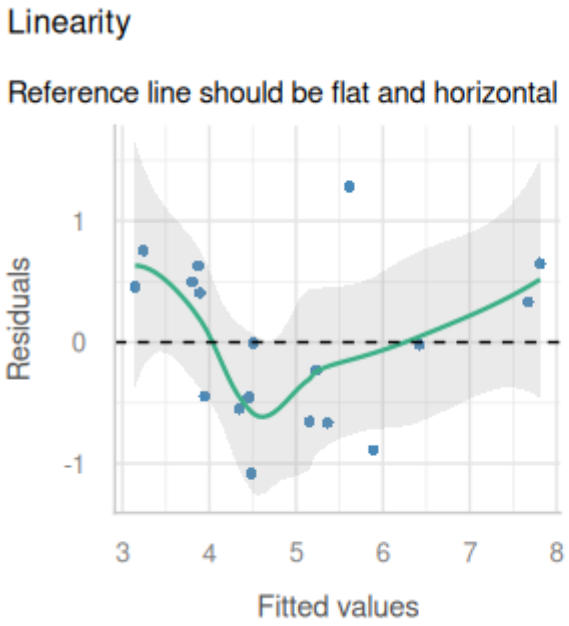
We only use the training data to fit the model. For this tutorial we will use the model we optimised last week:

Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogOther

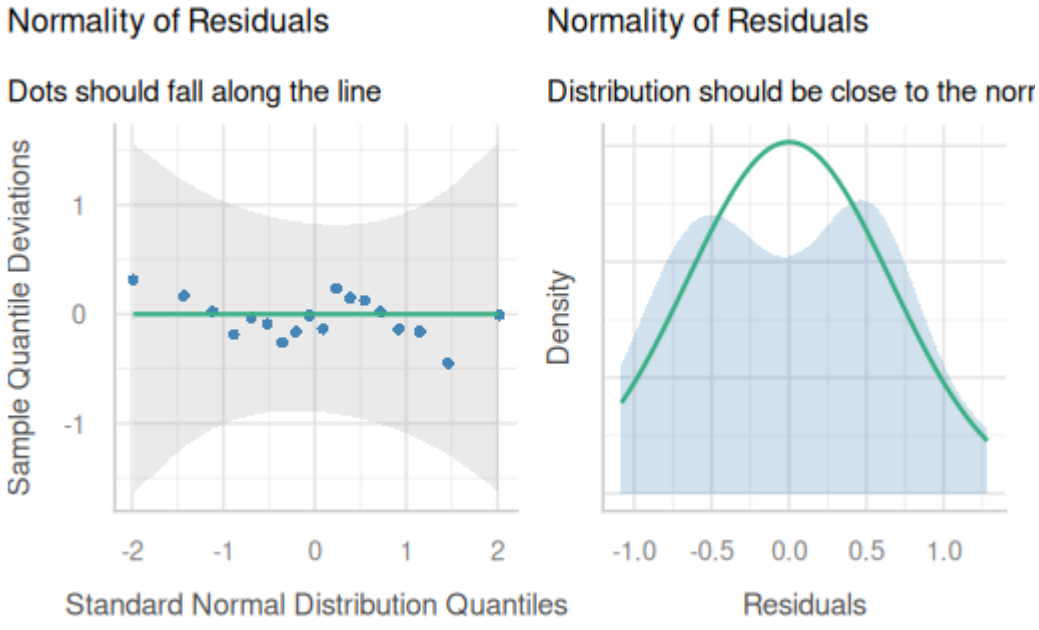
```
CODE
# fit the model to the training data

train_model <- lm(Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogOther, data =
dippers_train)
```

We can do a quick assumption check to ensure the model is appropriate for the training data, but we won't go through this in detail as we have already done this for the full dataset last week.

```
CODE
library(performance)
# Linearity
check_model(train_model, check = c("linearity"))
```

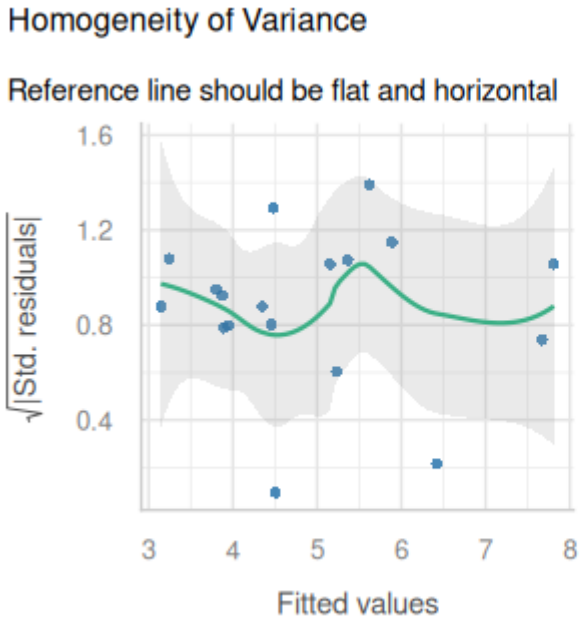



```
CODE
# Normality
check_model(train_model, check = c("qq", "normality"))
```

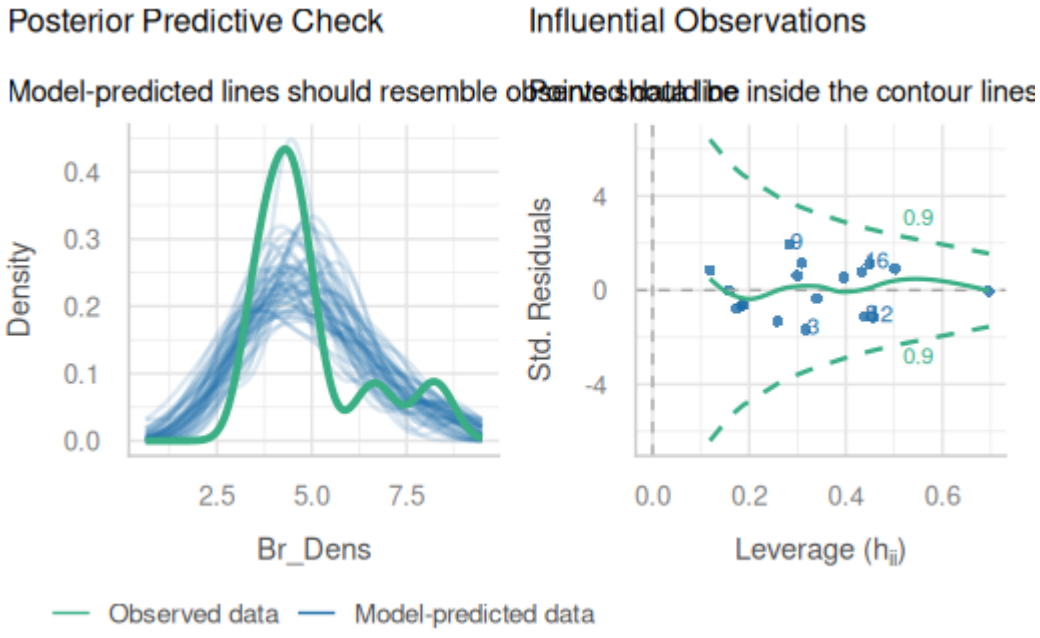



```
CODE
```

```
# Equal variance
check_model(train_model, check = c("homogeneity"))
```

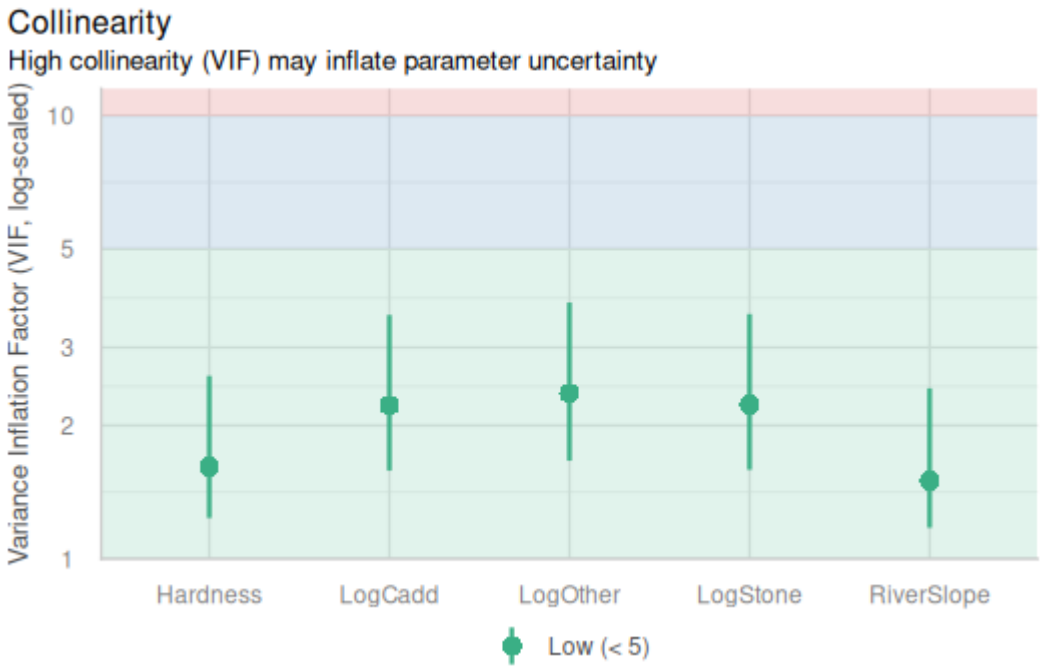



```
CODE
# Outliers
check_model(train_model, check = c("outliers", "pp_check"))
```



CODE

```
# VIF - visual
plot(performance::check_collinearity(train_model))
```



CODE

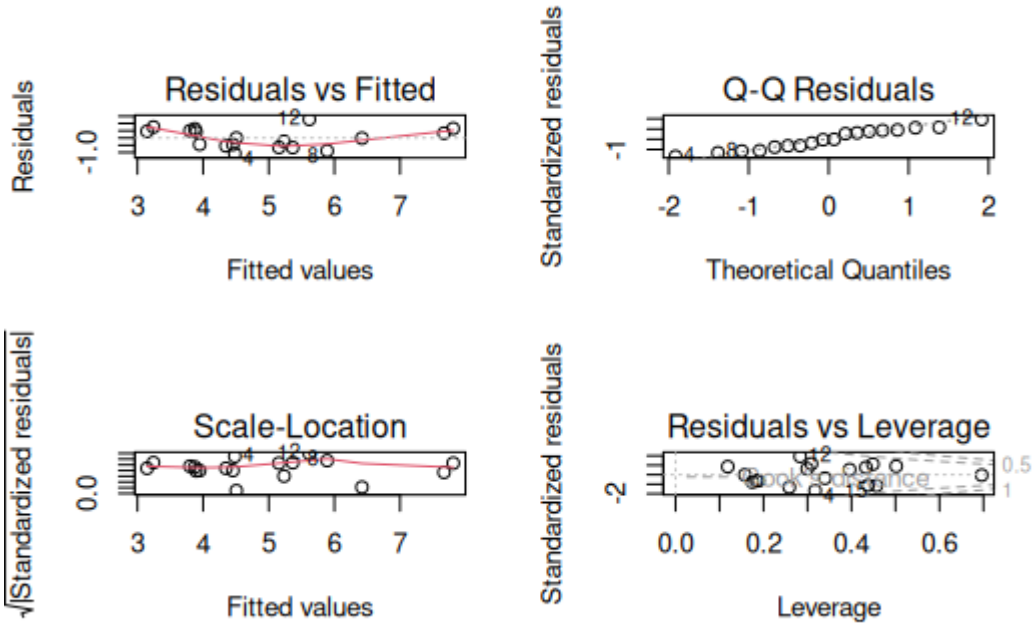
```
# VIF - numbers
# comes from the car package
car::vif(train_model)
```

OUTPUT

```
  Hardness RiverSlope    LogCadd   LogStone   LogOther 
  1.613588   1.500583   2.218855   2.229204   2.362211 
```

CODE

```
#### ALTERNATIVE: plot() function to check assumptions

par(mfrow=c(2,2))

plot(train_model)
```




```
CODE
par(mfrow=c(1,1))
```

- Linearity: The relationship between the predictors and the response variable appears to be potentially nonlinear, because although the points are scattered, the line is indicating a possible trend.
- Independence: The residuals appear to be independent.
- Normality: The residuals appear to be approximately normally distributed, as the points in the Q-Q plot roughly follow a straight line. When we look at the `check_model()` version of the plot we a slight n-shape to the points, but this is only small compared to the overall distribution of the points, so we are not too concerned about this.
- Equal variance: The residuals appear to have constant variance, as there is no clear pattern or fanning present in the standardised residuals vs fitted plot (scale-location in the base plot).
- Leverage: No single observations lying outside the Cook's distance line, indicating no influential observations.

We use the training set to fit the model and check assumptions. We can also use this subset to make predictions and check the quality of the predictions. The predictions made with this subset will often look quite good, but this is because the observed y value has been used in model fitting; therefore we do not assess the true predictive performance of the model using the training set alone. Instead we can compare the prediction quality statistics and plots with those of the test set to get an indication of overfitting.

The test subset is a set of values which are not used to fit the model, and is therefore unseen by the model. This allows us to pretend we have a new set of values, but then we have the advantage of the actual observed y variable sitting within the dataset too, so we are able to test how similar the predictions are to our real observation.

We will run through this using last week's data:

If the model is good, we expect the predictions to be close to the actual values.

If the model is bad, we expect the predictions to be far from the actual values.

The dataset can be obtained by:

Collecting new data.

Splitting the existing data into two parts before model building.

Data splitting

Cross-validation

use cars dataset?

This week's lecture introduced you to variable selection and how to produce a more parsimonious model. We explored developing our models by fitting the full model, checking assumptions and deciding whether anything needed transformation. Once we were happy assumptions were met we could consider whether all of our predictors were useful in explaining the variation in our response variable.

In this tutorial, we will explore a new dataset and follow the steps of model selection to produce a model that explains a sufficient amount of variation, without being unnecessarily complex.

You will learn how to:

1. Check for collinearity and influential observations in a MLR model
2. Run partial F-tests to compare nested models
3. Implement stepwise regression using the `step()` function in R

Setting up for our analysis

Workflow for fitting parsimonious models

1. Model development

- Start with full model and check assumptions (e.g. normality, homoscedasticity, linearity, etc.)
- Look for additional issues (e.g. multicollinearity, outliers, etc.) —> correlations, leverage plot, VIF plots and statistics
- Consider transformations (e.g. log, sqrt, etc.). –> Response: if assumptions aren't met, transform the response variable –> Predictors: it may help to explore transformations of skewed predictors before fitting the model to explore if there is a relationship present. Transforming predictors generally does not affect assumptions, but it can help to linearise relationships and make them easier to interpret.
- Following any transformations, test assumptions again.

2. Model selection

- Use VIF as an initial step to get rid of highly correlated predictors.
- Perform variable selection using backward elimination (good and fast), because:
- Using $r^2$2 as a criterion is not recommended (it is not a good measure of model fit, only a good measure of variance explained).
- Using partial F-test is good, but slow.

Packages

This tutorial uses `tidyverse`, `readxl`, `moments`, `psych`, `performance` and `car`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("tidyverse", "readxl", "moments", "psych", "performance", "car"))
```

If a package is already installed, R will simply reinstall it — no harm done.

The data

Figure 2: Cute little Dipper (bird) standing on a rock by the water. Image source: markmedcalf via Adobe Stock.

Dippers are thrush-sized birds living mainly in the upper reaches of rivers, which feed on benthic invertebrates by probing the river beds with their beaks. The dataset in this exercise contains data from a biological survey which examined the nature of the variables thought to influence the breeding of British dippers.

Twenty-two sites were included in the survey. Some of the variables have been transformed.

Data can be found in: **Dippers.csv**

The variables measured were:

- `Altitude` site altitude
- `Hardness` water hardness
- `RiverSlope` river-bed slope
- `LogCadd` the numbers of caddis fly larvae, transformed
- `LogStone` the numbers of stonefly larvae, transformed
- `LogMay` the numbers of mayfly larvae, transformed
- `LogOther` the numbers of all other invertebrates collected, transformed
- `Br_Dens` the number of breeding pairs of dippers per 10 km of river

In the analyses, the four invertebrate variables were transformed using a $log(x+1)$ transformation.

`Br_Dens` will be the response variable in our models.

Analysis

Model development

(i) Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots.

CODE

```
# read in the data
dippers <- read.csv("data/Dippers.csv")

library(tidyverse)
glimpse(dippers) # check the data
```

OUTPUT

```
Rows: 22
Columns: 8
$ Altitude   <int> 259, 198, 251, 184, 145, 145, 198, 160, 251, 159, 160, 145,…
$ Hardness   <dbl> 12.20, 22.00, 26.30, 22.50, 29.50, 39.90, 42.80, 59.60, 69.…
$ RiverSlope <dbl> 10.90, 14.70, 6.90, 4.60, 1.91, 5.00, 6.20, 14.30, 4.60, 3.…
$ LogCadd    <dbl> 2.303, 2.890, 3.784, 4.419, 3.219, 3.932, 3.664, 4.431, 3.7…
$ LogStone   <dbl> 5.242, 4.344, 5.231, 5.242, 3.829, 4.898, 4.357, 6.337, 5.4…
$ LogMay     <dbl> 0.000, 3.401, 5.826, 5.749, 5.509, 5.749, 5.371, 0.000, 4.8…
$ LogOther   <dbl> 1.386, 1.609, 1.386, 1.386, 1.099, 3.045, 1.386, 2.944, 2.5…
$ Br_Dens    <dbl> 3.60, 4.30, 3.80, 3.40, 3.80, 4.50, 4.30, 5.00, 4.50, 3.40,…
```

CODE

```
# Correlations and scatterplots all in one (+ histograms for each variable)
library(psych)
```

OUTPUT

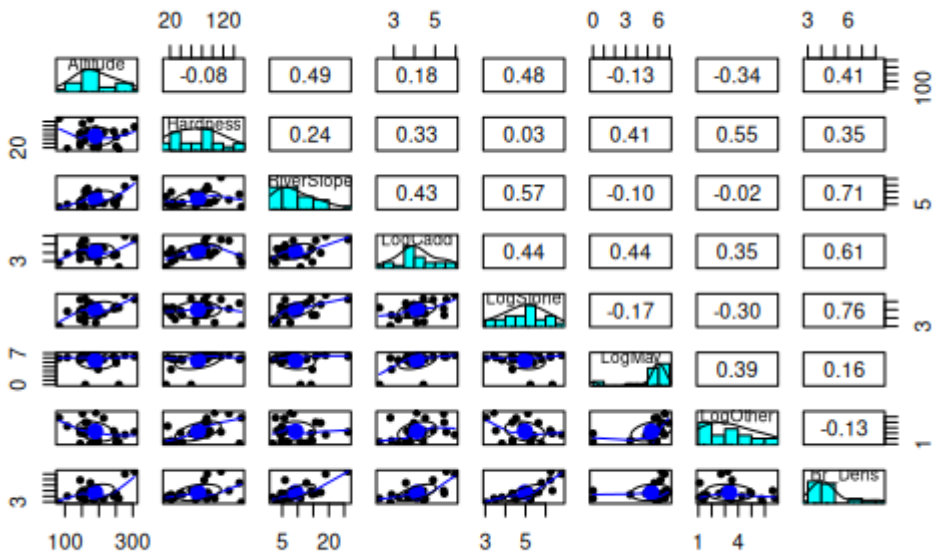
```

Attaching package: 'psych'
```

OUTPUT

```
The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

CODE

```
pairs.panels(dippers)
```



CODE

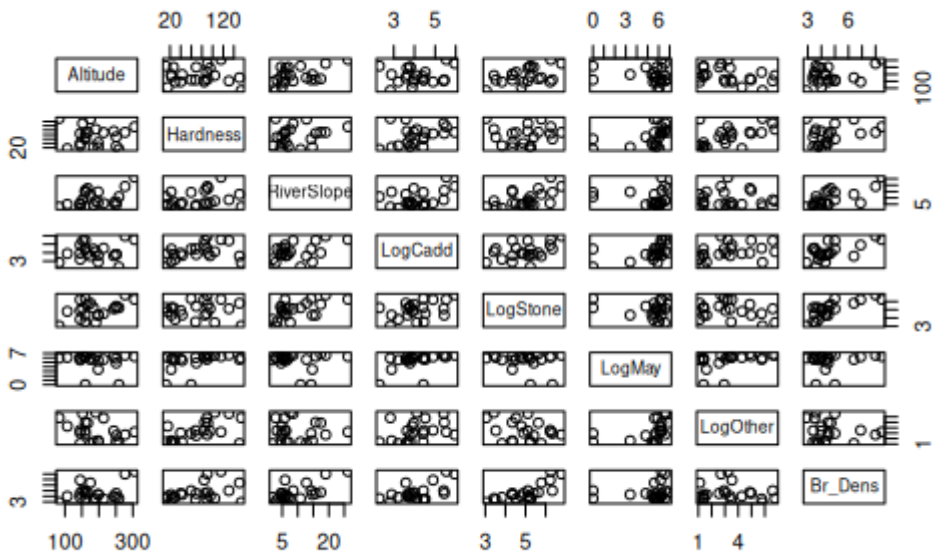
```
# can also get the same results separately
cor(dippers)
```

OUTPUT

```
              Altitude    Hardness  RiverSlope   LogCadd    LogStone
Altitude    1.00000000 -0.08170082  0.49003652 0.1807820  0.47736562
Hardness   -0.08170082  1.00000000  0.24208735 0.3337586  0.02951218
RiverSlope  0.49003652  0.24208735  1.00000000 0.4313081  0.57479242
LogCadd     0.18078205  0.33375857  0.43130806 1.0000000  0.44347440
LogStone    0.47736562  0.02951218  0.57479242 0.4434744  1.00000000
LogMay     -0.12782013  0.41356844 -0.09616686 0.4378978 -0.16744101
LogOther   -0.33859788  0.55399249 -0.02007183 0.3521440 -0.30025225
Br_Dens     0.40635349  0.35034817  0.71022164 0.6126891  0.76294367
                LogMay    LogOther    Br_Dens
Altitude   -0.12782013 -0.33859788  0.4063535
Hardness    0.41356844  0.55399249  0.3503482
RiverSlope -0.09616686 -0.02007183  0.7102216
LogCadd     0.43789780  0.35214400  0.6126891
LogStone   -0.16744101 -0.30025225  0.7629437
LogMay      1.00000000  0.38820063  0.1562659
LogOther    0.38820063  1.00000000 -0.1277591
Br_Dens     0.15626585 -0.12775913  1.0000000
```

CODE

```
pairs(dippers)
```



Between Br_Dens and other variables, there are a number of relatively stong correlations, particularly with LogStone (r = 763), RiverSlope (r = 0.710) and LogCadd (r = 0.613).

There are also some strong correlations between the predictor variables themselves, particularly between RiverSlope and LogStone (r = 0.575) and LogOther and Hardness (r = 0.554). This could be an issue for our model, as collinearity can make it difficult to interpret the model parameters and can inflate the standard errors of the estimates.

We can confirm if correlations between predictors may be an issue with the model by observing the Variance inflation factor (VIF) later on when we check our model assumptions.

(ii) Fit a multiple linear regression model to the data using all variables. Set `Br_Dens` as the response variable and the remaining variables as the predictors. This will be known as the 'full model'.

```
CODE
# fit the full model

full_model <- lm(Br_Dens ~ Altitude + Hardness + RiverSlope + LogCadd + LogStone + LogMay +
LogOther, data = dippers)
```

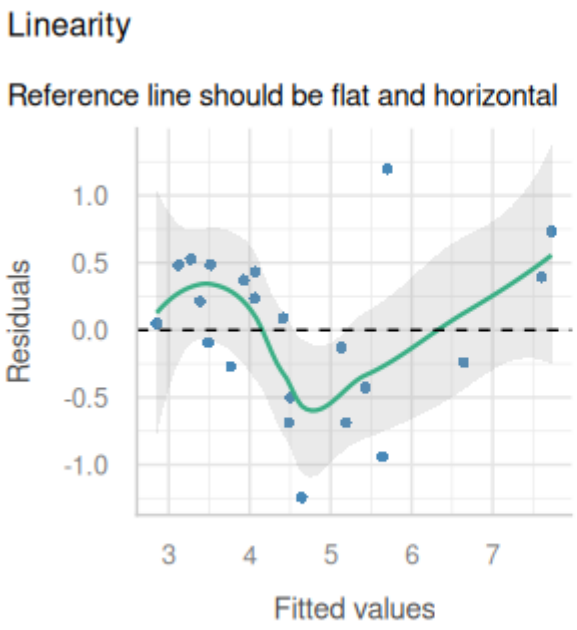
(iii) Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables?

The VIF will appear as one of the plots in the `check_model()` function, but it can also be helpful to get the values. You can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

The check_model function has been squishing plots, so it may be better at times to plot them individually. You may use the following code, adapted to your model object names:

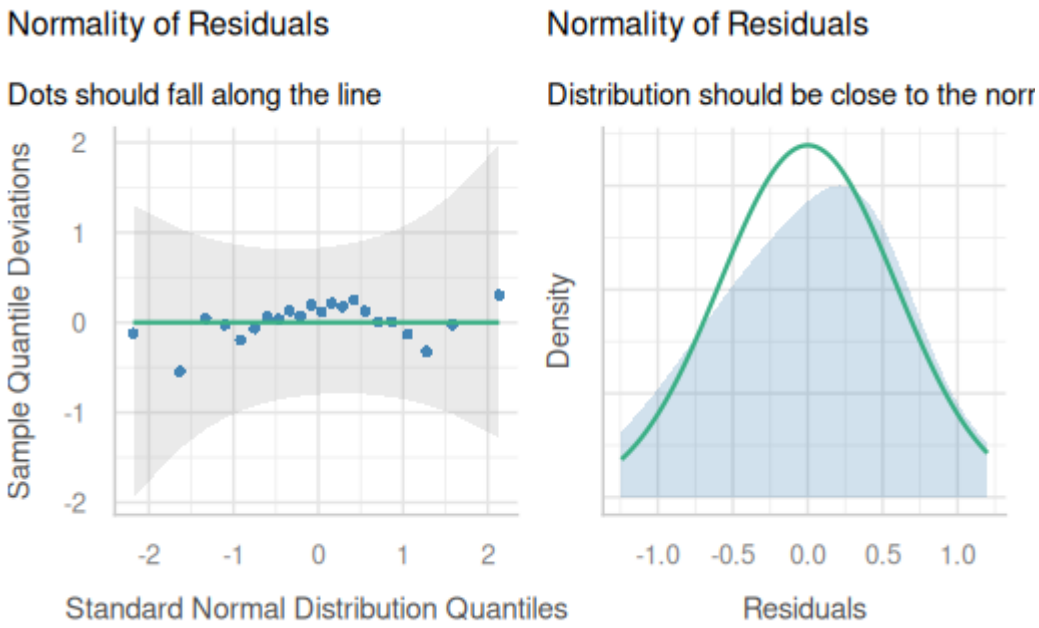
CODE

```
library(performance)

# Linearity
check_model(full_model, check = c("linearity"))
```

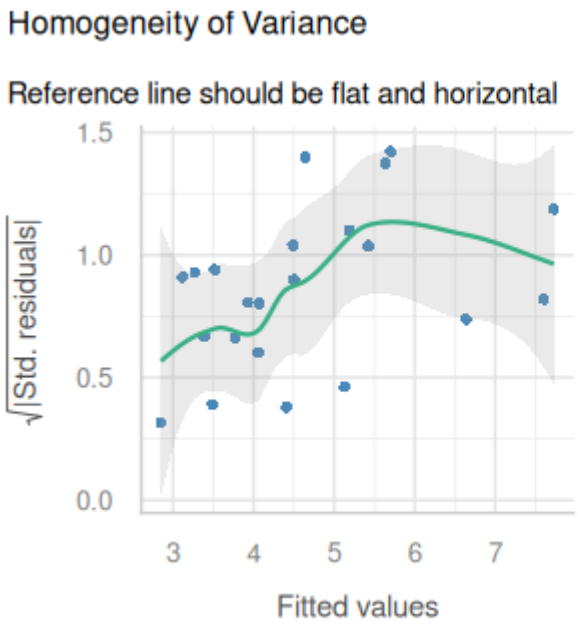


CODE

```
# Normality
check_model(full_model, check = c("qq", "normality"))
```

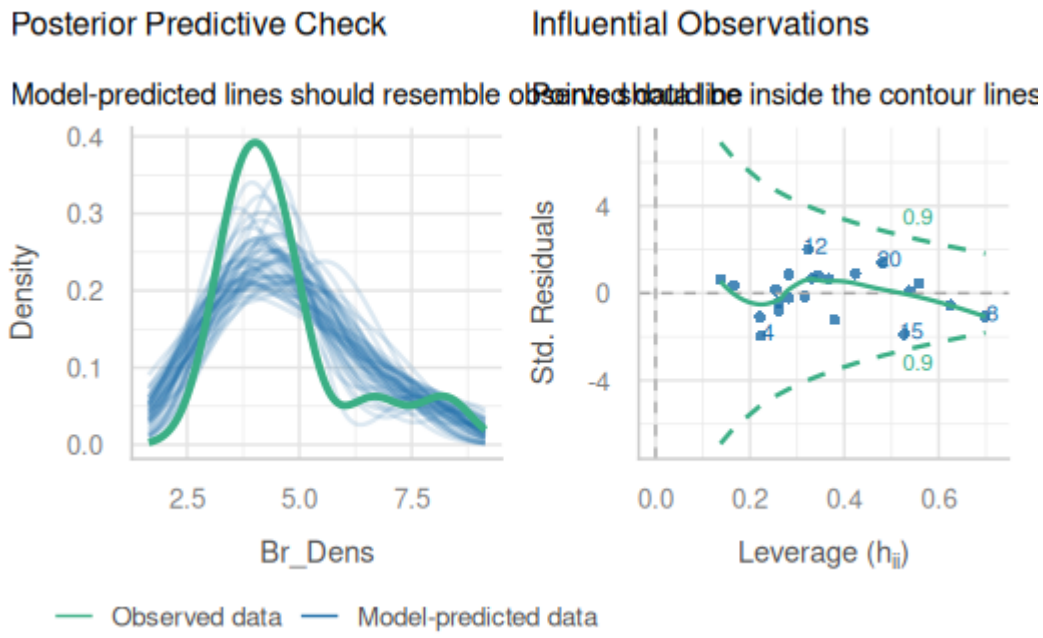



```
CODE
# Equal variance
check_model(full_model, check = c("homogeneity"))
```



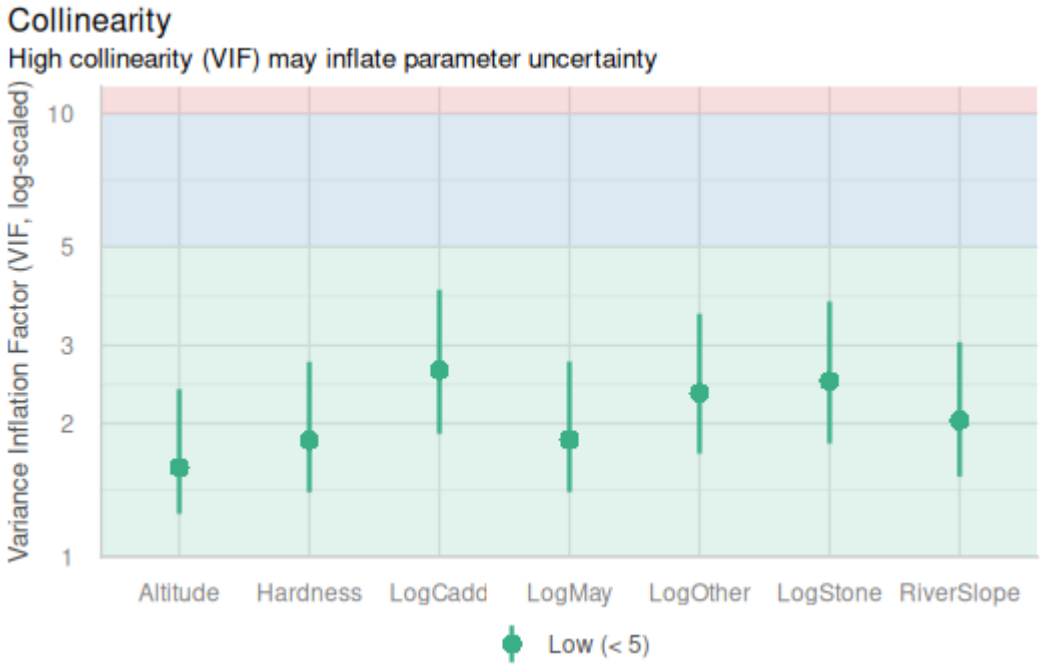

```
CODE
```

```
# Outliers
check_model(full_model, check = c("outliers", "pp_check"))
```



CODE

```
# VIF - visual
plot(performance::check_collinearity(full_model))
```



CODE

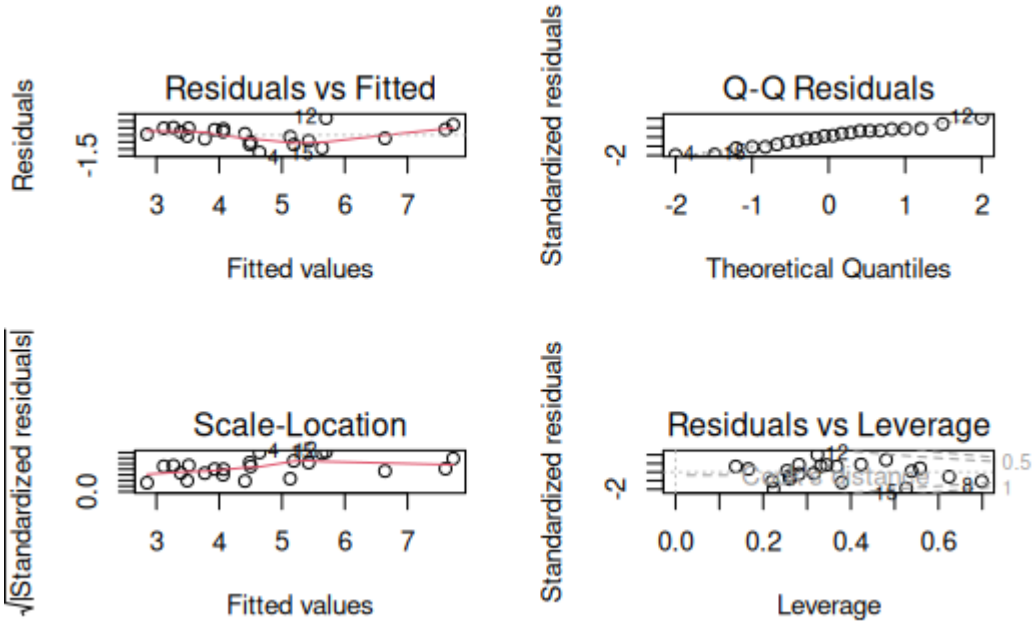
```
# VIF - numbers
# comes from the car package
car::vif(full_model)
```

OUTPUT

```
  Altitude   Hardness RiverSlope    LogCadd   LogStone     LogMay   LogOther
  1.589862   1.835610   2.030493   2.639160   2.489586   1.838431   2.341119
```

CODE

```
#### ALTERNATIVE: plot() function to check assumptions

par(mfrow=c(2,2))

plot(full_model)
```



CODE

```
par(mfrow=c(1,1))
```

- Linearity: The relationship between the predictors and the response variable appears to be linear, as the residuals are mostly scattered around the horizontal line at zero in the residuals vs fitted plot.
- Independence: The residuals appear to be independent, as there is no clear pattern in the residuals vs fitted plot.
- Normality: The residuals appear to be approximately normally distributed, as the points in the Q-Q plot roughly follow a straight line. When we look at the `check_model()` version

of the plot we a slight n-shape to the points, but this is only small compared to the overall distribution of the points, so we are not too concerned about this.

- Equal variance: The residuals appear to have constant variance, as there is no clear pattern or fanning present in the standardised residuals vs fitted plot (scale-location in the base plot).
- Leverage: No single observations lying outside the Cook's distance line, indicating no influential observations.

VIF and collinearity

VIF Thresholds vary, depending on who you ask. General rule is that if it is greater than 10, then you need to remove the associated predictor. Some conservative people prefer a value of 5, so we will go with this.

Our threshold is 5, so given all of our VIFs are below this, we are not too concerned about collinearity. However, because they are close to the threshold it is something to keep in mind when interpreting the model parameters.

(iv) Make a note of the model parameters and fit statistics for the full model.

CODE

```
summary(full_model)
```

OUTPUT

```

Call:
lm(formula = Br_Dens ~ Altitude + Hardness + RiverSlope + LogCadd +
    LogStone + LogMay + LogOther, data = dippers)

Residuals:
     Min       1Q   Median       3Q      Max
-1.24313 -0.38686  0.06925  0.42252  1.19902

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.881054   1.215055  -0.725   0.4803
Altitude    -0.001801   0.003537  -0.509   0.6186
Hardness     0.009956   0.004961   2.007   0.0645 .
RiverSlope   0.080254   0.035781   2.243   0.0416 *
LogCadd      0.408917   0.274412   1.490   0.1584
LogStone     0.604083   0.249207   2.424   0.0295 *
LogMay       0.114607   0.114116   1.004   0.3323
LogOther    -0.267073   0.131828  -2.026   0.0623 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7208 on 14 degrees of freedom
Multiple R-squared:  0.8433,    Adjusted R-squared:  0.765
F-statistic: 10.77 on 7 and 14 DF,  p-value: 0.0001085
```

We can see that there are two significant predictors (RiverSlope and LogStone) in the full model, but there are also a number of non-significant predictors. The adjusted R^2 value is 0.77, which indicates that the model explains 77% of the variation in the response variable.

Because there are so many non-significant predictors in the model, we may want to consider removing some of them to produce a more parsimonious model.

(v) Select the least significant predictor and remove it from the model.

In this case, the least significant predictor is `Altitude` (P = 0.619). Remove it from the model and run the model again. This will be known as the 'reduced model'.

CODE

```
reduced_model <- lm(Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogMay + LogOther, data
= dippers)
```

(vi) Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this?

CODE

```
summary(full_model)
```

OUTPUT

```

Call:
lm(formula = Br_Dens ~ Altitude + Hardness + RiverSlope + LogCadd +
    LogStone + LogMay + LogOther, data = dippers)

Residuals:
     Min       1Q   Median       3Q      Max
-1.24313 -0.38686  0.06925  0.42252  1.19902

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.881054   1.215055  -0.725   0.4803
Altitude    -0.001801   0.003537  -0.509   0.6186
Hardness     0.009956   0.004961   2.007   0.0645 .
RiverSlope   0.080254   0.035781   2.243   0.0416 *
LogCadd      0.408917   0.274412   1.490   0.1584
LogStone     0.604083   0.249207   2.424   0.0295 *
LogMay       0.114607   0.114116   1.004   0.3323
LogOther    -0.267073   0.131828  -2.026   0.0623 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7208 on 14 degrees of freedom
Multiple R-squared:  0.8433,    Adjusted R-squared:  0.765
F-statistic: 10.77 on 7 and 14 DF,  p-value: 0.0001085
```

CODE

```
summary(reduced_model)
```

OUTPUT

```

Call:
lm(formula = Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone +
    LogMay + LogOther, data = dippers)

Residuals:
     Min       1Q   Median       3Q      Max
-1.21925 -0.36896  0.03995  0.40149  1.27830
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.112573   1.098551  -1.013   0.3272
Hardness     0.010101   0.004829   2.092   0.0539 .
RiverSlope   0.073829   0.032643   2.262   0.0390 *
LogCadd      0.405503   0.267470   1.516   0.1503
LogStone     0.587737   0.240949   2.439   0.0276 *
LogMay       0.111440   0.111096   1.003   0.3317
LogOther    -0.251481   0.125014  -2.012   0.0626 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7028 on 15 degrees of freedom
Multiple R-squared:  0.8404,    Adjusted R-squared:  0.7766
F-statistic: 13.17 on 6 and 15 DF,  p-value: 3.139e-05
```

Removing the `Altitude` predictor from the model has slightly increased the adjusted R^2 value. This is an improvement in terms of parsimony, but we will need to use hypothesis testing to determine whether the reduced model is a *significant* improvement on the full model. We can do this using an F-test.

> **i** Checkpoint
>
> You should now be able to prepare an appropriate model by refining it, checking your assumptions, and ensuring collinearity is not an issue.

Model selection with F-tests

We wil use the partial F-test to determine whether there is a significant difference between the models.

(i) State hypotheses

H0: no significant difference between the full and reduced models H1: There is a significant difference between the full and reduced models

If we fail to reject H0, then we would opt for the reduced model as it is more parsimonious (i.e. it is explaining a similar amount of variation with fewer predictors).

If we reject H0, then we would opt for the full model as it is a significant improvement on the reduced model.

(ii) Run the test

Once we have established our hypotheses, we can perform the F-test using the `anova()` function to compare the full and reduced models.

```
CODE
```

```
anova(full_model, reduced_model)
```

OUTPUT

```
Analysis of Variance Table

Model 1: Br_Dens ~ Altitude + Hardness + RiverSlope + LogCadd + LogStone +
    LogMay + LogOther
Model 2: Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogMay +
    LogOther
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     14 7.2739
2     15 7.4085 -1  -0.13464 0.2591 0.6186
```

(iii) Are the models significantly different?

The P-value is 0.6186, which is larger than 0.05, so we fail to reject the null hypothesis.

This means that the full model is not a significant improvement on the reduced model, and we would opt for the reduced model as it is more parsimonious.

i Checkpoint

You should now be able to use an F-test to determine whether the reduced model is a significant improvement on the full model.

Remember: the F-test is only useful if comparing nested models.

Model selection using Stepwise regression

When we have many predictors, this manual testing can become time consuming, so we can choose to use the `step()` function in R, which performs stepwise regression. This function will automatically remove the least significant predictor and test the model again, until it finds the best model based on the AIC criterion.

(i) Run the `step()` function on the full model to perform stepwise regression.

CODE

```
step_model <- step(full_model, direction = "backward")
```

OUTPUT

```
Start:  AIC=-8.35
Br_Dens ~ Altitude + Hardness + RiverSlope + LogCadd + LogStone +
    LogMay + LogOther

             Df Sum of Sq     RSS     AIC
- Altitude    1   0.13464  7.4085 -9.9451
```

```
- LogMay      1   0.52404  7.7979 -8.8181
<none>                    7.2739 -8.3486
- LogCadd     1   1.15372  8.4276 -7.1097
- Hardness    1   2.09230  9.3662 -4.7866
- LogOther    1   2.13246  9.4063 -4.6925
- RiverSlope  1   2.61383  9.8877 -3.5945
- LogStone    1   3.05288 10.3268 -2.6387

Step:  AIC=-9.95
Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogMay +
    LogOther

             Df Sum of Sq     RSS      AIC
- LogMay      1   0.49696  7.9055 -10.5167
<none>                    7.4085  -9.9451
- LogCadd     1   1.13522  8.5437  -8.8086
- LogOther    1   1.99863  9.4071  -6.6906
- Hardness    1   2.16061  9.5691  -6.3150
- RiverSlope  1   2.52643  9.9349  -5.4897
- LogStone    1   2.93869 10.3472  -4.5952

Step:  AIC=-10.52
Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogOther

             Df Sum of Sq     RSS      AIC
<none>                    7.9055 -10.5167
- RiverSlope  1    2.1227 10.0282  -7.2842
- LogOther    1    2.3128 10.2182  -6.8712
- LogStone    1    2.4742 10.3797  -6.5262
- LogCadd     1    2.9091 10.8146  -5.6232
- Hardness    1    3.3747 11.2801  -4.6960
```

(ii) What is the final model that the `step()` function has produced? How does it compare to the full and reduced models?

Since we have stored the step() output as an object, we can just call the object, which will show us the final model:

CODE

```
step_model
```

OUTPUT

```

Call:
lm(formula = Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone +
    LogOther, data = dippers)

Coefficients:
(Intercept)     Hardness   RiverSlope      LogCadd     LogStone     LogOther
   -0.72462      0.01181      0.06538      0.54880      0.51249     -0.26813
```

CODE

```
## OR

summary(step_model)
```

OUTPUT

```
Call:
lm(formula = Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone +
    LogOther, data = dippers)

Residuals:
     Min       1Q   Median       3Q      Max
-1.18190 -0.49120 -0.08373  0.41721  1.31073

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.724619   1.028416  -0.705   0.4912
Hardness     0.011811   0.004519   2.613   0.0188 *
RiverSlope   0.065384   0.031545   2.073   0.0547 .
LogCadd      0.548799   0.226169   2.426   0.0274 *
LogStone     0.512494   0.229020   2.238   0.0398 *
LogOther    -0.268128   0.123932  -2.164   0.0460 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7029 on 16 degrees of freedom
Multiple R-squared:  0.8297,    Adjusted R-squared:  0.7765
F-statistic: 15.59 on 5 and 16 DF,  p-value: 1.169e-05
```

We can see that the selected model is:

Br_Dens ~ Hardness + RiverSlope + LogCadd + LogStone + LogOther

Or with the addition of the partial regression coefficients, our equation is:

Br_Dens = −0.725 + 0.012 * Hardness + 0.065 * RiverSlope + 0.549 * LogCadd + 0.512 * LogStone - 0.268 * LogOther

We can see there has also been a marginal change in the adjusted R^2 value, which is now 0.7765, compared to 0.765 for the full model and 0.7766 for the reduced model. This supports the idea that the full model explained a similar amount of variation in the response compared to the simplest model obtained from the `step()` function. Therefore it makes sense to select the simpler model.

(iii) How does the AIC of the stepwise regression model compare to the full and reduced models?

We can test this using the AIC() function:

CODE

```
AIC(full_model, reduced_model, step_model)
```

OUTPUT

```
              df      AIC
full_model     9 56.08470
reduced_model  8 54.48820
step_model     7 53.91655
```

Full_model AIC = 56.1

Reduced_model AIC = 54.5

step_model AIC = 53.9

As we select the model with the smallest AIC, we can confirm that the model selected by the `step()` function is better.

You may notice that the AIC values are different between the `step()` function and the `AIC()` function. This is because the `step()` function uses a different method to calculate the AIC values, which is based on the likelihood of the model, while the `AIC()` function uses a different method based on the residual sum of squares.

> **i** Checkpoint
>
> You should now be able to run the `step()` function to select the most appropriate model.

Wrap-up

In this tutorial, we used the dippers dataset to get some practice testing model assumptions and checking for collinearity. We then used partial F-tests and stepwise regression to select a more parsimonious model.

Example exam questions

1. What assumptions do we need to check for when fitting a multiple linear regression?
2. Why don't we just rely on the adjusted r^2 to compare models?
3. Can F-tests be run on non-nested models? Why or why not?